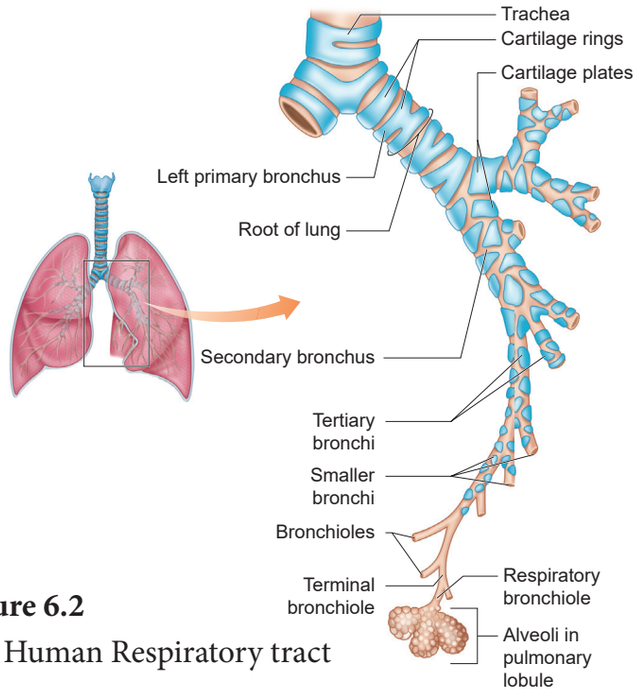


Figure 6.2
The Human Respiratory tract

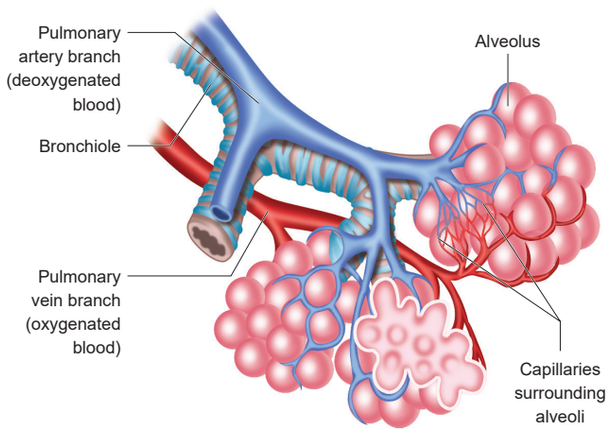


Figure 6.3 Structure of alveoli

The steps involved in respiration are

i. The exchange of air between the atmosphere and the lungs.

ii. The exchange of O_2 and CO_2 between the lungs and the blood.

iii. Transport of O_2 and CO_2 by the blood.

iv. Exchange of gases between the blood and the cells.

v. Uptake of O_2 by the cells for various activities and the release of CO_2.



SURFACTANTS are the thin non–cellular films made of protein and phospholipids covering the alveolar membrane. The surfactant lowers the surface tension in the alveoli and prevents the lungs from collapsing. It also prevents pulmonary oedema. Premature Babies have low levels of surfactant in the alveoli may develop the new born respiratory distress syndrome (NRDS) because the synthesis of surfactants begins only after the 25th week of gestation.

6.3 Mechanism of breathing

The movement of air between the atmosphere and the lungs is known as ventilation or breathing. Inspiration and expiration are the two phases of breathing. Inspiration is the movement of atmospheric air into the lungs and expiration is the movement of alveolar air that diffuse out of the lungs. (Figure 6.4)

Lungs do not contain muscle fibres but expands and contracts by the movement of the ribs and diaphragm. The diaphragm is a sheet of tissue which separates the thorax from the abdomen. In a relaxed state, the diaphragm is domed shaped. Ribs are moved by the intercostal muscles. External and internal intercostal muscles found between the ribs and the diaphragm helps in creating pressure gradients. Inspiration occurs if the pressure inside the lungs (intrapulmonary pressure) is less than the atmospheric pressure likewise expiration takes place when the pressure within the lungs is higher than the atmospheric pressure.

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